THE BLOWING UP IS FINAL AMONG MORPHISMS MAKING THE INVERSE IMAGE OF Z CARTIER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The statement carries an optional argument 'Universal property blowing up' in the environment header; it is part of the counted statement span and is preserved verbatim.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The blowing up makes the inverse image of Z an effective Cartier divisor. Lemma lemma-blowing-up-gives-effective-Cartier-divisor (8382-8395), used at proof lines 8442-8443. Places the blowing up inside the subcategory.
- Relative Proj and maps of graded algebras give morphisms. Constructions, Section constructions-section-relative-proj, used at proof lines 8453-8456. Produces the morphism Y -> X' from the triple (1, f, psi).
- The complement of an effective Cartier divisor is scheme theoretically dense. Lemma lemma-complement-effective-Cartier-divisor, used at proof lines 8460-8461. Is what makes the morphism unique.
- Equality of morphisms on a scheme theoretically dense open. Morphisms, Lemma morphisms-lemma-equality-of-morphisms and Constructions, Lemma constructions-lemma-relative-proj-separated, used at proof lines 8461-8466. Turns the uniqueness on Y \ D into uniqueness on Y.

Source reference. Original lemma 01O2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2179–2179. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 01NS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2169–2169. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02OS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3527–3527. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07ZU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10159–10159. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01RH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2302–2302. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2 (Universal property blowing up)**.** *Let X be a scheme. Let $Z \subset X$ be a closed subscheme. Let $\mathcal{C}$ be the full subcategory of (Sch/X) consisting of $Y \to X$ such that the inverse image of Z is an effective Cartier divisor on Y. Then the blowing up $b : X' \to X$ of Z in X is a final object of $\mathcal{C}$.*

Proof. We see that $b : X' \to X$ is an object of $\mathcal{C}$ according to Lemma 02OS. Let $f : Y \to X$ be an object of $\mathcal{C}$. We have to show there exists a unique morphism $Y \to X'$ over X. Let $D = f^{-1}(Z)$. Let $\mathcal{I} \subset \mathcal{O}_X$ be the ideal sheaf of Z and let $\mathcal{I}_D$ be the ideal sheaf of D. Then $f^*\mathcal{I} \to \mathcal{I}_D$ is a surjection to an invertible $\mathcal{O}_Y$-module. This extends to a map $\psi : \bigoplus f^*\mathcal{I}^d \to \bigoplus \mathcal{I}_D^d$ of graded $\mathcal{O}_Y$-algebras. (We observe that $\mathcal{I}_D^d = \mathcal{I}_D^{\otimes d}$ as D is an effective Cartier divisor.) By the material in Constructions, Section 01NS the triple $(1, f : Y \to X, \psi)$ defines a morphism $Y \to X'$ over X. The restriction

$$Y \setminus D \longrightarrow X' \setminus b^{-1}(Z) = X \setminus Z$$

is unique. The open $Y \setminus D$ is scheme theoretically dense in Y according to Lemma 07ZU. Thus the morphism $Y \to X'$ is unique by Morphisms, Lemma 01RH (also b is separated by Constructions, Lemma 01O2). □